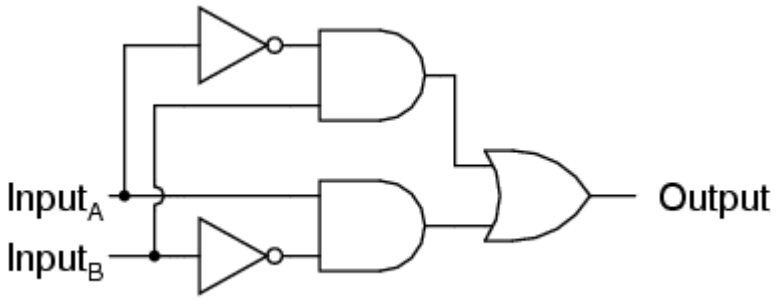
Tutorial 03 - Logic Gates

1. Draw the NOT (Inverter) Gate symbol and its truth table.
2. Draw the AND Gate symbol and its truth table.
3. Draw the OR Gate symbol and its truth table.
4. Draw the NAND Gate symbol and its truth table.
5. Draw the NOR Gate symbol and its truth table.
6. Draw the XOR Gate symbol and its truth table.
7. Draw the XNOR Gate symbol and its truth table.
8. Write a logical expression to denote the output of an AND gate.
9. Write a logical expression to denote the output of an OR gate.
10. What is meant by the complement of a logical value?
11. What is meant by a ‘Unary Operation’?
12. Name four Multiple Input Logic Gates.
13. Name four Derived Logic Gates

Challenge Questions (Optional but MUST try!)

1. Consider the following circuit diagram. Fill in the output column of the truth table provided based on different combinations of inputs.



A	B	Output
0	0	0
0	1	1
1	0	
1	1	

2. The output of a logic gate is ‘1’ when all its inputs are at logic 0. The gate is either:
    a. NAND or an XOR gate
    b. NOR or an XNOR gate
    c. an OR or an XNOR gate
    d. an AND or an XOR gate

3. How do you make a NOR gate out of a NAND gate using inverters (NOT gates)?
    a. Invert the output from the NAND gate
    b. Invert both the inputs to the NAND gate
    c. Invert one of the inputs to the NAND gate
    d. Invert both the inputs and output of the NAND gate

4. How do you make an AND gate out of an NAND gate using inverters (NOT gates)?
    a. Invert the output from the NAND gate
    b. Invert both the inputs to the NAND gate
    c. Invert one of the inputs to the NAND gate
    d. Invert both the inputs and output of the NAND gate

5. The output of an AND gate with three inputs, A, B, and C, is HIGH when _____.
    a. A = 1, B = 1, C = 0
    b. A = 0, B = 0, C = 0
    c. A = 1, B = 1, C = 1
    d. A = 1, B = 0, C = 1